

Digital Signal Processing

Final Project

Part I) ECG Signal Denoising for Telemedicine Applications

Objective

Design and implement digital filters to remove noise from Electrocardiogram (ECG) signals used in remote health monitoring and wearable medical systems.

Signal Characteristics

Typical ECG properties:

- Sampling frequency: 360 Hz
- Useful ECG bandwidth: 0.5 – 100 Hz

Common noise components:

- Baseline wander (< 0.5 Hz) due to respiration
- Power-line interference at 50 Hz
- Muscle (EMG) noise between 20 – 150 Hz

ECG Data Source

You are required to use ECG recordings from the publicly available **MIT-BIH Arrhythmia Database** hosted on PhysioNet.

The database contains 48 annotated ECG recordings sampled at 360 Hz and widely used in biomedical signal processing research.

You may download the database from: <https://physionet.org/content/mitdb/1.0.0/>

You are encouraged to use the following records:

- **Record 100** (primarily **normal** sinus rhythm)
- **Record 106** (contains **abnormal** ventricular ectopic beats)

Each record includes three files:

- .dat – ECG signal samples
- .hea – signal header information
- .atr – beat annotations

You can extract a segment of the signal for filter design, analysis, and visualization.

Program requirements:

1) Problem Definition and Specifications

You are required to:

1. Describe ECG signal characteristics & clearly define the filtering objectives
2. Identify the required filter types:
 - High-pass filter (baseline removal)
 - Notch filter at 50 Hz
 - Low-pass filter (muscle noise reduction)
3. Define and justify filter specifications:
 - Passband ripple
 - Stopband attenuation
 - Transition bandwidth
 - Cutoff frequencies
4. Identify realistic implementation constraints such as:
 - Maximum filter order
 - Computational complexity
 - Latency
 - Phase distortion limitations

2) Filter Design and Analysis

You must design at least three different digital filters, for example:

- FIR (Window-based method)
- IIR Butterworth
- IIR Chebyshev (Type I or II)

For each design you must:

- Specify filter type and design method
- Provide filter coefficients
- Plot magnitude and phase responses
- Plot impulse and step responses
- Show pole-zero diagram
- Verify compliance with specifications

3) Implementation and Validation

- Apply the designed filters to ECG data.
- Compare the ECG signal before and after filtering.
- Demonstrate effectiveness using:
 - Power spectral density (Welch method)
 - Spectrogram (STFT)
 - Quantitative indicators such as SNR improvement
- Provide time-domain waveform comparison.
- Discuss distortion and possible limitations.

Part II) Multi-Band Speech Equalizer for Podcast Enhancement

Objective

Develop a MATLAB-based multi-band digital equalizer for speech clarity enhancement in podcasts, recorded lectures, and voice recordings.

Program Requirements:

The program must allow the user to input:

- Audio file name
- Gain of each frequency band (in dB)
- Filter type (FIR or IIR)
- Filter order
- Output sample rate

Modes of Operation

1- Preset Mode (Speech-Optimized Bands)

Predefined frequency bands:

- 0–100 Hz
- 100–300 Hz
- 300–800 Hz
- 800–2 kHz
- 2–5 kHz
- 5–10 kHz
- 10–20 kHz

2- Custom Mode

- Minimum 5 bands
- Maximum 10 bands
- First band must start at 0 Hz
- Last band must end at 20 kHz

Filter Design Options

You may design the equalizer filters using either FIR or IIR structures.

FIR Filters

- Window types:
 - Hamming
 - Hanning
 - Blackman
- User-defined or default filter order

IIR Filters

- Filter types:
 - Butterworth
 - Chebyshev Type I
 - Chebyshev Type II
- User-defined or default filter order

Filter Design and Analysis

You must generate for each band:

- Magnitude response
- Phase response
- Impulse response
- Step response
- Filter order
- Pole-zero plot

Signal Processing Steps

1. Design band filters.
2. Filter the input signal.
3. Apply gain (in **dB**).
4. Combine filtered bands in the time domain.
5. Compare original and processed signals in time & frequency domains

6. Plot Power Spectral Density.
7. Generate spectrogram.
8. Play and save the output file.
9. Demonstrate:
 - FIR implementation
 - IIR implementation
 - Output sample rate multiplied by four
 - Output sample rate reduced to half

Submission requirements:

1. Each group must consist of a maximum of (3) students.
2. Both programs must be implemented.
3. Copied codes or reports will receive a **ZERO**.
4. **Deliverables:** Submit a single PDF report including:

Part I)

- a) All required analysis, filter design comparisons, specification justification, performance evaluation, and validation results.
- b) MATLAB codes of all problems.
- c) Figures requested. Label your figures properly.

Part II)

- a) Compare spectra before and after equalization, Analyze spectrogram differences, Perform listening evaluation, and Discuss artifacts, distortion, and trade-offs
- b) MATLAB codes of all problems
- c) Figures requested. Label your figures properly.